

Mixed multiplicity in general dimension:

$$Z_{m+2}(n) \sim \left(\prod k_i^{-1} \right) b^{k_0(q-p)} \frac{C(p, q)}{2^m m!} n^{-p/2} (\log n)^m$$

Claude, with Daniel Murfet

2026-09-06

Abstract

The first mixed case in arbitrary dimension: the innermost coordinate carries a candidate exponent q strictly larger than the common exponent p of the remaining $m + 1$ coordinates. The chart integral reduces exactly to $(\prod k_i^{-1}) c^{-p} b^{k_0(q-p)} H_m(cb^{\sum k_i})$, where H is the divide-and-integrate tower of unit 49 built on the *noncritical base* $G_1(y) = \int_0^y x^{p-q-1} F_{q-1}(x) dx$. That base is admissible with mass $C(p, q)$ (unit 46's constant), tail rate $\varepsilon = q - p$ (which is why unit 51 was needed) and $\delta = p$, so the expansion machinery applies verbatim and yields the asymptotic in the title: multiplicity $m + 1$, the noncritical coordinate contributing only the constant $b^{k_0(q-p)} C(p, q)$. Unit 40 is the case $k = (1, 1, 1)$, $h = (1, 0, 0)$ up to the order of integration. Zero `sorry/axiom`.

1 Setting

After the innermost coordinate, $Z_1(c) = k_0^{-1} c^{-q} F_{q-1}(cb^{k_0})$ (unit 50's reduction with $p := q$). The first minimal coordinate turns c^{-q} into c^{-p} through the substitution $\int_0^{b^{k_1}} s^{p-1} (cs)^{-q} F_{q-1}(\dots) ds$, whose kernel $s^{p-q-1} F_{q-1}$ is exactly the integrand of $C(p, q)$; the scaling lemma produces G_1 and the factor $(cb^{k_0})^{q-p}$. Every further minimal coordinate is one divide-and-integrate step, identical to the all-equal case.

2 The things formalised

```
theorem iterChartGen_step ... (hpd : ((h d : ) + 1) / k d = p)
  (hZ : c, 0 < c →
    iterChartGen a b k h d c = K * c ^ (-p) * D * iterDivPrim G j (c * B))
  (c : ) (hc : 0 < c) :
    iterChartGen a b k h (d + 1) c
      = K * (1 / (k d : )) * c ^ (-p) * D * iterDivPrim G (j + 1) (c * (B * b ^ (k d)))
noncomputable def noLogBase ( a p q y : ) : :=
  x in Ioc 0 y, x ^ (p - q - 1) * weightedPrimitive a (q - 1) x
theorem noLogBase_logBase ( a p q : ) (h : 0 < ) (hp : 0 < p) (hlt : p < q) :
  LogBase (noLogBase a p q) (noLogConst a p q) (weightedMass a (q - 1) / (q - p))
    (Real.exp ( * a ^ 2 / 2) / (p * q)) p (q - p)
theorem iterChartGen_mixed_eq ... :
  iterChartGen a b k h (m + 2) c
    = ( i Finset.range (m + 2), (1 : ) / k i) * c ^ (-p) * b ^ ((k 0 : ) * (q - p))
      * iterDivPrim (noLogBase a p q) m (c * b ^ ( i Finset.range (m + 2), k i))
```

```

theorem iterChartGen_mixed_isEquivalent ... (hm : 0 < m) ... (hlt : p < q) :
  (fun n => iterChartGen a b k h (m + 2) (Real.sqrt n)) ~[atTop]
    fun n => ( i Finset.range (m + 2), (1 : ) / k i) * b ^ ((k 0 : ) * (q - p))
      * (noLogConst a p q / (2 ^ m * (m.factorial : )))
      * (n ^ (-(p / 2)) * Real.log n ^ m)

```

3 Proof strategy

The recursion step at a minimal coordinate is factored out (`iterChartGen_step`) for an arbitrary tower $H = \text{iterDivPrim } G$ and arbitrary constants K, D, B ; it is unit 50’s inductive step with the base abstracted. The mixed reduction is an induction on m whose base case does the two innermost coordinates by hand and whose step is one application of the step lemma. The base G_1 is monotone (hence measurable), bounded by $C(p, q)$ (its tail is a nonnegative integral), bounded near 0 by $(E/(pq))y^p$ via $F_{q-1} \leq Ex^q/q$, and has tail $\int_L^\infty x^{p-q-1} F_{q-1} \leq A_{q-1} L^{-(q-p)}/(q-p)$. With these, the asymptotic is unit 50’s assembly with $A := C(p, q)$ and the extra constant $b^{k_0(q-p)}$. The reduction was checked by three-dimensional quadrature in two configurations before formalising.

4 Roadblocks and resolutions

- `Real.rpow_natCast b (k 0)` rewrote every b^{k_0} , including the one inside the integration bound, so a later rewrite failed to match; state the single needed identity $(b^{k_0})^{q-p} = b^{k_0(q-p)}$ as a `have` and rewrite with it.
- `Finset.prod_range_succ` without arguments fired on the wrong product (the left-hand side’s `range (m+2)` rather than the right-hand side’s `range (m+2+1)`); pass the function and index explicitly.

5 What was Mathlib, what was new

Mathlib: `integral_Ioi_rpow_of_lt`, `integrableOn_Ioi_rpow_of_lt`, `setIntegral_mono_set`, `Monotone.measurable`. New: the generic recursion step, the noncritical base and its `LogBase` certificate, and the mixed multiplicity theorem.

6 Lessons

Two abstractions from earlier units (`LogBase` with a rate, and the step lemma) made the mixed case a matter of exhibiting one new base function and verifying four bounds. The pattern “noncritical coordinates produce a bounded base; minimal coordinates produce logarithms” is now visible as a theorem rather than as three worked examples.

7 Outcome

The leading-term theory of `thm:TaylorTree` in the chart geometry now covers: all exponents equal in any dimension with arbitrary (k, h) (unit 50), one noncritical inner coordinate plus any number of minimal ones (this unit), and all three regimes in two dimensions (units 44–48). Remaining at this level: several noncritical coordinates (whose iterated bases have polynomial growth, not bounded), and the paper-form product-set statements.